REPORT

GLAUCOMA

Vitamin B_3 modulates mitochondrial vulnerability and prevents glaucoma in aged mice

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Glaucomas are neurodegenerative diseases that cause vision loss, especially in the elderly. The mechanisms initiating glaucoma and driving neuronal vulnerability during normal aging are unknown. Studying glaucoma-prone mice, we show that mitochondrial abnormalities are an early driver of neuronal dysfunction, occurring before detectable degeneration. Retinal levels of nicotinamide adenine dinucleotide (NAD^+), a key molecule in energy and redox metabolism) decrease with age and render aging neurons vulnerable to disease-related insults. Oral administration of the NAD^+ precursor nicotinamide (vitamin B_3), and/or gene therapy (driving expression of *Nmnat1*, a key NAD^+-producing enzyme), was protective both prophylactically and as an intervention. At the highest dose tested, 93% of eyes did not develop glaucoma. This supports therapeutic use of vitamin B_3 in glaucoma and potentially other age-related neurodegenerations.

Glaucoma is a group of complex, multifactorial diseases characterized by the progressive dysfunction and loss of retinal ganglion cells (RGCs), leading to vision loss. Glaucoma is one of the most common neurodegenerative diseases worldwide, affecting more than 70 million people (*1*). High intraocular pressure (IOP) and increasing age are important risk factors for glaucoma (*2*, *3*). However, specific mechanisms rendering RGCs more vulnerable to damage with age are unknown. Here, we address how increasing age and high IOP interact to drive neurodegeneration using DBA/2J (D2) mice, a widely used model of chronic, age-related, inherited glaucoma (*4*).

We used RNA-sequencing (RNA-seq) to elucidate age and IOP-dependent molecular changes within RGCs that precede glaucomatous neurodegeneration. We analyzed RGCs of 9-month-old D2 mice [in a stage termed early glaucoma—high IOP and molecular changes but lacking neurodegeneration (*2*)]; 4-month-old D2 mice (in a stage preceding high IOP); and age-, sex-, and strain-matched D2-*Gpnmb*$^+$ controls [which do not develop high IOP or glaucoma (*4*)] (Fig. 1). RGCs were isolated (fig. S1), and their RNA was sequenced at a depth of 35 million reads per sample. Unsupervised hierarchical clustering (HC) allowed molecular definition of early glaucoma stages among samples that were still morphologically indistinguishable from age-matched D2-*Gpnmb*$^+$ or young controls. HC identified four distinct groups of 9-month-old D2 samples (groups 1 to 4). Group 1 clustered with all of the control samples and represents D2 RGCs with no detectable glaucoma at a molecular level. Although groups 2 to 4 were all early stages, increasing group number reflects greater glaucoma progression at a transcriptomic level (Fig. 1A and fig. S2, A and B). As disease progressed, there was an increase in transcript abundance that was most pronounced for mitochondrial reads (Fig. 1B). Emerging evidence suggests that imbalances in the relative proportions of mitochondrial proteins encoded by nuclear and mitochondrial genomes negatively affect mitochondrial function (*5*). In D2 groups 2 to 4, differential expression of genes encoding mitochondrial proteins, as well as significant enrichment [false discovery rate (FDR) < 0.05] of differentially expressed (DE) genes in the mitochondrial dysfunction and oxidative phosphorylation pathways, further point to mitochondrial abnormalities (Fig. 1, B to G; fig. S2, C to G; and tables S1 to S3). Pathway analysis identified enrichment of eukaryotic initiation factor 2 (eIF2) and mammalian target of rapamycin (mTOR) signaling transcripts (Fig. 1C). eIF2 is a key regulator of redox homeostasis and cellular adaptations to stress and was the most enriched pathway in group 2 (first distinguishable stage from controls). Through mTOR inhibition, it promotes survival in the presence of oxidative stress (*6*–*8*) and likely protects from mitochondrial abnormalities in RGCs at this early disease stage. Mitochondrial fission (Fig. 1F) and mitochondrial unfolded protein response genes were also DE (Fig. 1G) (*9*–*11*). Electron microscopy (EM) revealed abnormal mitochondria with reduced cristae volume in the dendrites of D2 RGCs but not in those of control RGCs (Fig. 1, H and I). These mitochondrial EM findings coincide with synapse loss in 9-month-old D2 retinas (*12*), with early decreases in pattern electroretinogram amplitude (PERG) (*13*) and an increase in retinal cytochrome c levels (fig. S2, D and F). Extending previous studies (*14*, *15*), our data demonstrate that mitochondrial perturbations are among the very first changes occurring within RGCs during glaucoma.

Guided by the above data, we assessed metabolites in retinas with increasing age and disease (D2 and D2-*Gpnmb*$^+$ at 4, 9, and 12 months). We detected early decreases in metabolites that are central to healthy mitochondrial metabolism and protection from oxidative stress: nicotinamide adenine dinucleotide (NAD^+) and the reduced form of NAD^+ (NADH) [total NAD, NAD(t)] and both the oxidized and reduced forms of glutathione [total glutathione, glutathione(t)] (Fig. 2A and fig. S2, H and I). These age-dependent decreases were not a response to IOP insult(s), as they also occurred in control D2-*Gpnmb*$^+$ retinas (fig. S2I). These decreases are expected to sensitize retinal neurons to disease-related stresses and mitochondrial dysfunction. There are increased levels of hypoxia-inducible factor HIF-1α mRNA and protein [a key metabolic regulator during perturbed redox states (*16*)] in the ganglion cell layer early in glaucoma, which suggests that there is greater metabolic stress in RGCs than in other retinal neurons (fig. S3, A and B). Our data suggest that RGCs go through a period of mitochondrial stress and metabolite depletion, which potentially moves them toward fatty acid metabolism (fig. S4). Fatty acid β-oxidation can increase generation of free radicals and/or reactive oxygen species (ROS) (*17*). Both RNA-seq (fig. S2C) and γ-H2AX immunostaining (fig. S2, J and K) results support increased ROS and DNA damage within RGCs early in glaucoma. Providing a link between DNA damage and increased metabolic stress, poly(ADP-ribose) polymerase (PARP) activity (NAD consuming) is induced in RGCs with age (fig. S5, A and B).

Our data support a model where age-dependent declines of NAD^+ and glutathione in the retina render RGCs vulnerable to damage from elevated IOP. Thus, increasing NAD levels would be predicted to protect IOP-insulted eyes from glaucomatous changes by decreasing the probability of metabolic and/or energetic failure and rendering the RGCs more resilient to IOP-induced stress. Oral supplementation of vitamin B_3 [nicotinamide (NAM), a precursor of NAD] has been successfully used to correct disturbances in NAD^+ metabolism in two mouse models of preeclampsia (*18*). Accordingly, we administered NAM to D2 mice, initially at the same dose (550 mg/kilogram of body weight per day, NAM^{Lo}) (Fig. 2). NAM administration in drinking water prevented the decline of NAD levels up to 12 months of age (a standard end stage for assessing neurodegeneration in this glaucoma model) (Fig. 2A). The finding that NAM^{Lo} did not alter IOP (fig. S6) but protected from glaucoma supported our neuronal vulnerability hypothesis.

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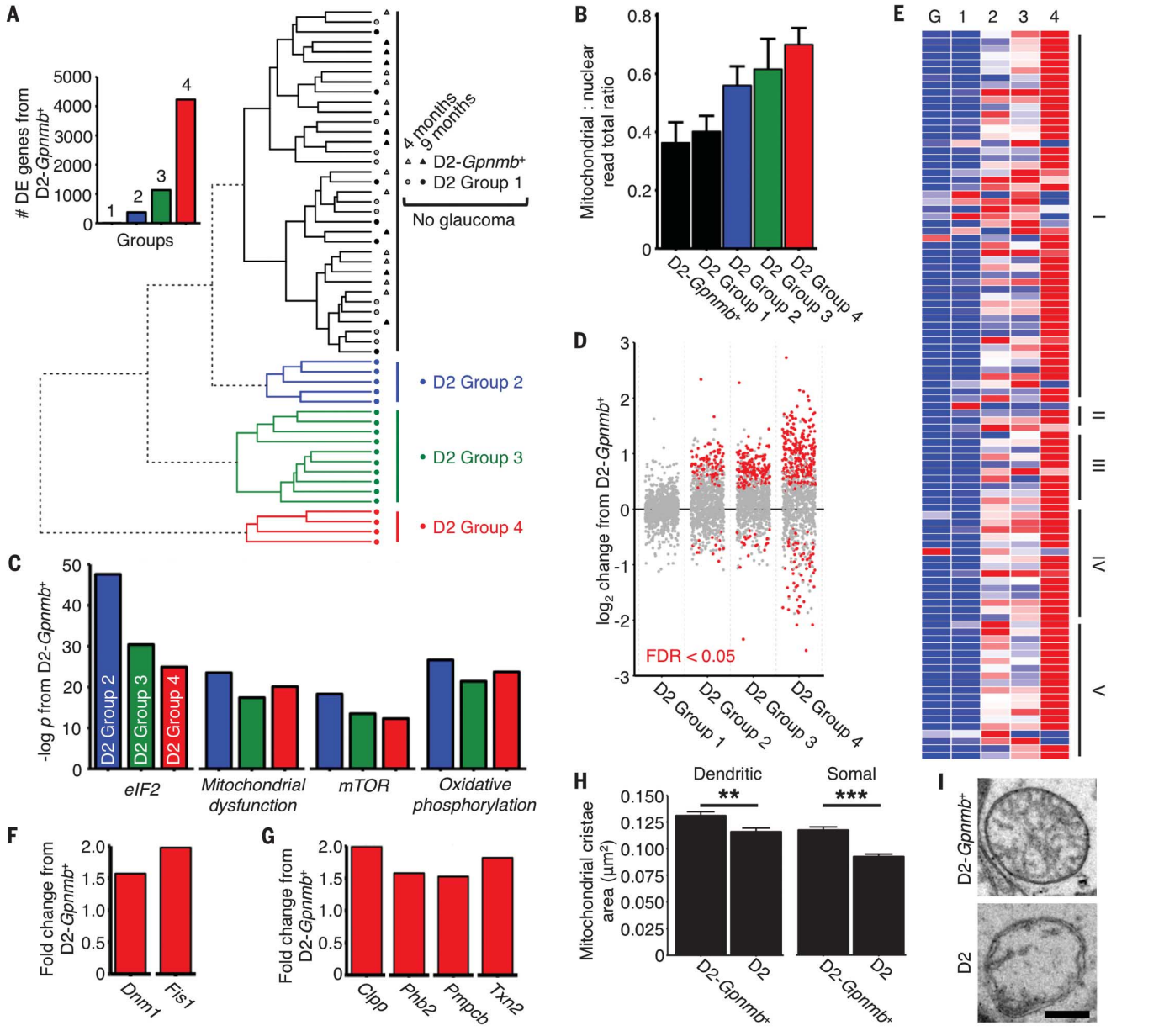


Fig. 1. Mitochondrial dysfunction is associated with progressive RGC damage in glaucoma. (**A**) RGC samples were divided into molecularly distinct groups by hierarchical clustering (HC) of RNA-seq–determined gene expression. Control (D2-*Gpnmb*$^+$) and young samples were molecularly similar (Spearman's rho; $n = 63$ samples; each sample was derived from a different mouse). Samples from D2-*Gpnmb*$^+$ RGCs, triangles; samples from D2 RGCs, circles. (Inset) Number of DE genes ($q < 0.05$) between D2-*Gpnmb*$^+$ and each group. D2 group 1 and D2-*Gpnmb*$^+$ represent no glaucoma at a molecular or transcriptomic level. Transcriptomic sequencing was performed twice, and batch correction was performed using RUVSeq (remove unwanted variation from RNA-seq data, see Materials and Methods). (**B**) Mitochondrial:nuclear read total ratio increased with greater HC distance from controls. Colors match key in (A). Error bars represent SEM. (**C**) Significantly enriched pathways between clusters and control based on Ingenuity Pathway Analysis (there are no differentially expressed pathways in D2 group 1) (see also table S3). (**D**) Transcript expression primarily increased for nuclear encoded mitochondrial proteins with increasing HC distance from controls. Dots represent individual genes; gray, not differentially expressed; red, differentially expressed at $q < 0.05$. Genes taken from mouse MitoCarta2.0 (*28*). (**E**) Oxidative phosphorylation genes were differentially expressed across all groups. Red, highest expression; blue, lowest expression; I to V, mitochondrial complexes I to V (tabulated in table S1); G, D2-*Gpnmb*$^+$; 1 to 4, D2 groups 1 to 4. (**F**) RNA-seq identified increased mitochondrial fission gene transcripts early in glaucoma and (**G**) suggests an early mitochondrial unfolded protein response compared with that of controls. Data shown are for D2 group 4. (**H** and **I**) Mitochondria of D2 mice have decreased cristae volume and decreased cristae:total volume ratio (not shown) in RGC somal and dendritic mitochondria (there was no significant difference in total mitochondrial size or volume) ($n > 400$ mitochondria from six retinas per group; pooled data are shown). Scale bar, 350 nm. All data are for mice at 9 months of age unless otherwise stated. $^{**}P < 0.01$; $^{***}P < 0.001$ (Student's t test). See also tables S1 and S2.

NAM was protective both prophylactically (starting at 6 months, before IOP elevation in most eyes in our colony) and interventionally (starting at 9 months, when the majority of eyes have had continuing IOP elevation) (Fig. 2B). NAM significantly reduced the incidence of optic nerve degeneration (Fig. 2, B and E), prevented RGC soma loss and retinal nerve fiber layer thinning (Fig. 2, C and E), and protected visual function, as

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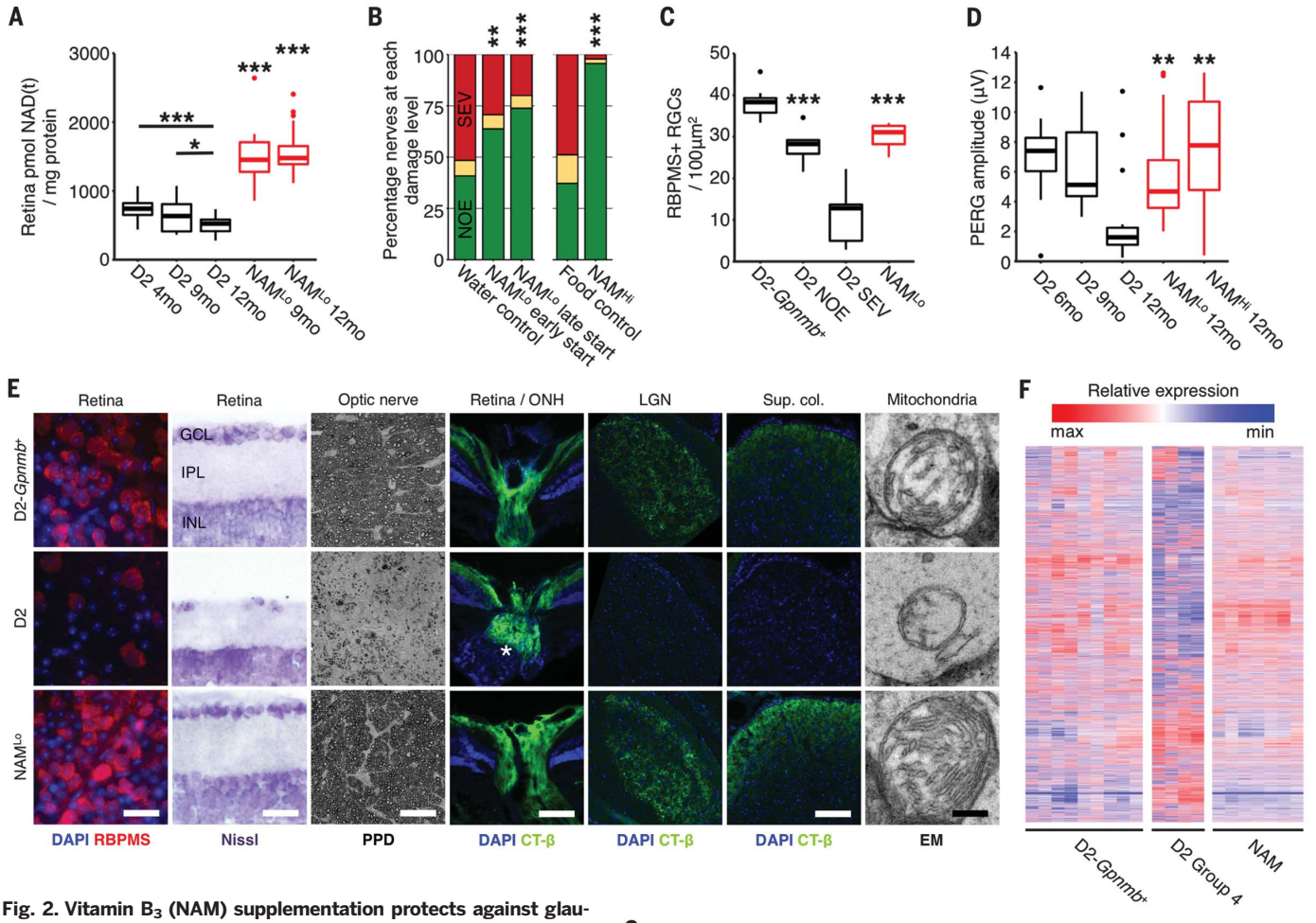


Fig. 2. Vitamin B_3 (NAM) supplementation protects against glaucoma development in mice. (**A**) NAD(t) levels were increased in NAM-treated D2 retinas as measured by colorimetric assay (n = 22 per group). Mo, months. (**B** and **E**) NAM intervention protected from optic nerve degeneration as assessed by PPD staining paraphenylenediamine, a sensitive stain for damaged axons). Green, no or early damage [<5% axon loss; no or early (NOE)]; yellow, moderate damage (~30% axon loss; MOD); red, severe (>50% axon loss; SEV) damage. Early start indicates mice that started treatment at 6 months (before IOP elevation in most eyes in our colony and, thus, prophylactic). Late start indicates mice that started treatment at 9 months (when the majority of eyes have had continuing IOP elevation, thus interventional). $^{**}P < 0.01$; $^{***}P < 0.001$ (Fisher's exact test). (**C** and **E**) NAM protected from RGC soma loss [number of somas positive for RNA binding proteins with multiple splicing (RBPMS+ cells), n = 8 per group]; the density drop between D2 and D2-*Gpnmb*$^+$ is due to pressure-induced stretching. (**D**) NAM protected from early loss in PERG amplitude (n > 20 per group). (E) NAM protected from RGC soma loss (n = 8 per group), retinal nerve fiber layer and inner plexiform layer (IPL) thinning (n = 8 per group), optic nerve degeneration (n > 50 per group), and loss of anterograde axoplasmic transport (n = 20 per group). DAPI, 4′,6-diamidino-2-phenylindole; GCL, ganglion cell layer; INL, inner nuclear layer Corresponding markers and color keys are beneath each column. Scale bars: RBPMS (a specific marker of RGCs; immunofluorescence), 20 μm; Nissl (a pan-neuronal stain; light microscopy), 20 μm; PPD (light microscopy), 20 μm; CT-β, 100 μm (for retina; immunofluorescence); 200 μm (for LGN and Sup. col.). ONH, optic nerve head; LGN, lateral geniculate nucleus; Sup. col., superior colliculus. White asterisk denotes loss of axonal transport at the site of the ONH. (**F**) Heat map of gene expression (all expressed genes) shows that NAM-treated RCGs were molecularly similar to controls. (**G**) Individual gene expression plots show metabolic and DNA damage pathways were returned to normal in NAM-treated RGCs. Dots represent individual genes; gray, not differentially expressed; red, differentially expressed at $q < 0.05$ compared with the D2-*Gpnmb*$^+$ 9-month-old control. (A), (C), and (D): $^*P < 0.05$; $^{**}P < 0.01$; $^{***}P < 0.001$ (Student's t test). For box plots, center hinge represents the mean and the upper and lower hinges represent the first and third quartiles; whiskers represent 1.5 times the interquartile range; values beyond the whiskers are plotted as outliers. See also fig. S4 and tables S1 to S3.

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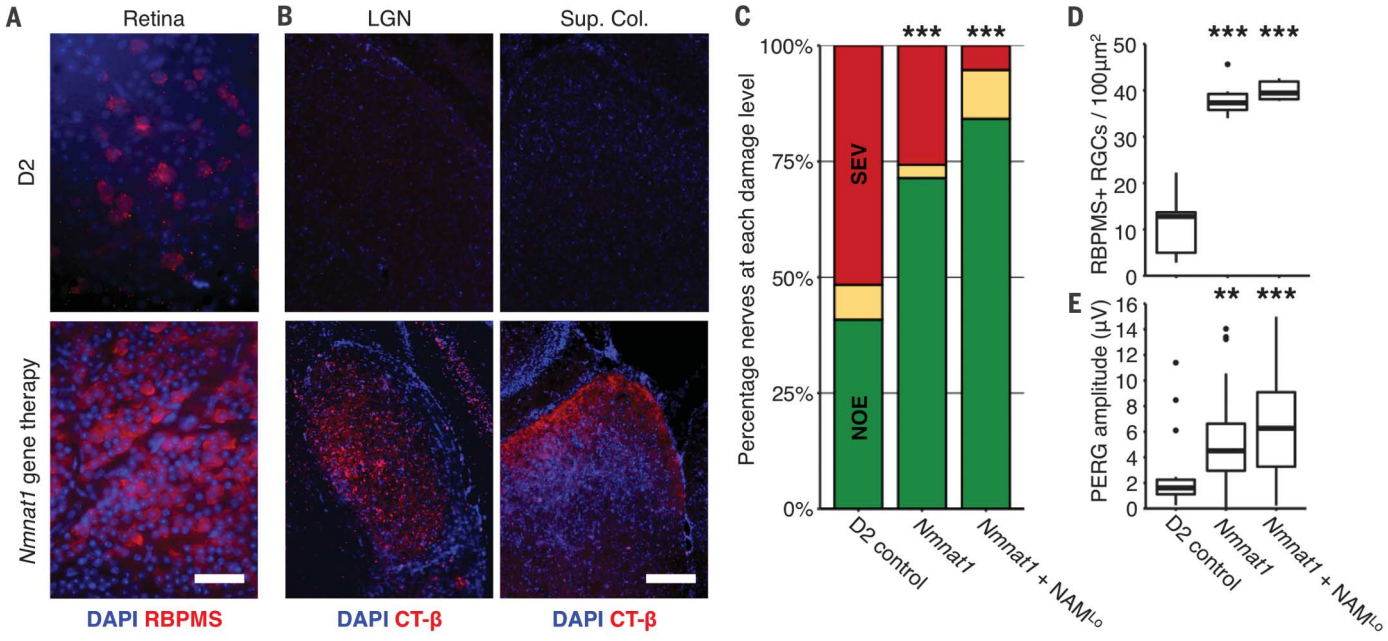


Fig. 3. Gene therapy protected eyes from glaucomatous neuron degeneration. D2 eyes were intravitreally injected at 5.5 months with the adeno-associated virus AAV2.2 carrying a plasmid to overexpress murine *Nmnat1* under a CMV promoter. (**A**) *Nmnat1* overexpression prevented RGC soma loss (red) (scale bar, 50 µm) and loss of anterograde axoplasmic transport (n = 10 per group) (as demonstrated in Fig. 2). (**B**) Soma loss (red). Scale bar, 100 µm. LGN, lateral geniculate nucleus; Sup. Col., superior colliculus. *Nmnat1* gene therapy also protected D2 eyes with elevated IOP against (**C**) optic nerve degeneration (red; n > 40 per group; ***P < 0.001, Fisher's exact test); (**D**) soma loss (n = 6 per group); and (**E**) PERG amplitude (n > 20 per group). Addition of NAM^{LO} in drinking water afforded additional protection against optic nerve degeneration (*Nmnat1* compared to *Nmnat1* + NAM^{LO} = P < 0.001, Fisher's exact test). (C), (D), and (E): **P < 0.01; ***P < 0.001 (Student's t test).

assessed by PERG (*13*) (Fig. 2D and fig. S3E). NAM prevented RGC axonal loss, and these axons continued to support anterograde axonal transport (Fig. 2E). NAM administration was sufficient to inhibit the formation of dysfunctional mitochondria with abnormal cristae (Fig. 2E and fig. S3, G and H) and also limited synapse loss that occurs in this model (*12*) (fig. S3, C and D). Lipid droplet formation was also prevented in aged D2 retinas (fig. S3F). NAM also decreased PARP activation and limited levels of DNA damage and transcriptional induction of HIF-1α (fig. S3, A and B), all of which reflected less-perturbed cellular metabolism. NAM prevented even the earliest molecular signs of glaucoma in most treated eyes, as assessed by RNA-seq (Fig. 2, F and G, and fig. S7), and prevented the majority of age-related gene expression changes within RGCs (fig. S8). This highlights the unexpected potency of NAM in decreasing metabolic disruption and prevention of glaucoma.

Attempting to further decrease the probability of glaucoma, we administered a higher dose of NAM (2000 mg/kg per day, NAM^{Hi}). NAM^{Hi} was extremely protective, with 93% of treated eyes having no optic nerve damage (Fig. 2B). The degree of protection afforded by administering this single molecule is unprecedented and unanticipated. Although NAM^{Lo} demonstrates a clear neuroprotective effect (no effect on IOP), NAM^{Hi} lessens the degree of IOP elevation (fig. S6). This indicates that NAM can protect against age-related pathogenic processes in other cell types in addition to RGCs (*11*, *16*, *19*). Therefore, NAM, a single molecule that protects against both IOP elevation and neural vulnerability, may have great potential for glaucoma treatment; however, human studies are needed.

NMNAT2 (nicotinamide/nicotinic acid mononucleotide adenylyltransferase 2) is emerging as an important NAD-producing enzyme in axons, and protects from axon degeneration (*20*). Ongoing stress negatively impacts *Nmnat2* expression in RGCs (q < 0.05 in D2 group 4) (fig. S7F). This decline of NMNAT2 may induce vulnerability to axon degeneration in glaucoma. *NMNAT2* expression is decreased in brains with Alzheimer's disease and is highly variable in aged postmortem human brains (*21*). Such variation in expression may contribute to individual differences in vulnerability to various neurodegenerations.

Glaucoma is a complex disease involving multiple insults. Mechanical axon damage and local inflammation are two important contributors in glaucoma (*22*–*24*). To more fully assess the general effectiveness of NAM treatment, we tested its efficacy in two models of RGC death that are used to model these glaucomatous insults. We used a tissue culture model of axotomy and intravitreal injections of soluble murine tumor necrosis factor–α (TNFα), which can drive local inflammation as well as mitochondrial dysfunction and is implicated in glaucoma (*25*). NAM robustly protected cultured retinas from RGC somal degeneration (fig. S9, A and B). NAM also protected against a loss of PERG amplitude and cell loss in TNFα-injected eyes (fig. S9, C to E). Given these protections against severe acute insults, NAM could have broad implications for treating glaucoma and potentially other age-related neurodegenerative diseases.

Gene therapy is an attractive approach for overcoming compliance issues and improving efficacy. In the eye, gene therapy has proven successful for rare human Mendelian disorders (*26*). Viral gene therapy allows a large number of cells to be transfected potentially lifelong by delivering a targeted gene product. To date, gene therapy has not been successfully applied to complex human diseases. Given that age is a common risk factor for most glaucoma, protecting from age-related declines in NAD may generally protect many glaucoma patients. Thus, we sought to support NAD^+-producing cellular machinery through the overexpression of *Nmnat1*, a terminal enzyme in NAD^+ production, to further test our NAD hypothesis. We chose this approach as we reasoned that using NAM phosphoribosyltransferase, the rate-limiting enzyme in NAD synthesis, may pose complications due to its cytokine functions and overproduction of NAM mononucleotide, which may participate in axon degeneration (*27*). D2 eyes were injected once with AAV2.2 containing the *Nmnat1* gene and a gene for green fluorescent protein (*GFP*) under a cytomegalovirus (CMV) promoter expressed as a single transcript (Fig. 3). *Nmnat1* expression (as assessed by GFP expression) was robust in RGCs 2 weeks after injection (expressed in >83% of RGCs) and remained robust

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through to the end-stage time point (12 months) (fig. S10). Overexpression of *Nmnat1* was sufficient to prevent axon and soma loss (Fig. 3, A to D), to preserve axoplasmic transport (Fig. 3B), and to preserve electrical activity in RGCs (PERG) (Fig. 3E). Glaucomatous nerve damage was absent in >70% of treated eyes. Because NMNAT1 catalyzes the terminal step in NAD production, the major protective effects of NAM treatment likely result from driving NAD production in neurons rather than other NAD-independent mechanisms (but partial contributions from other mechanisms cannot be completely excluded). We further assessed the effects of combinational therapy of *Nmnat1* and NAM^{Lo}. This combination afforded significant additional protection over *Nmnat1* or NAM^{Lo} alone, with 84% of eyes having no detectable glaucoma (~4-fold decreased risk of developing glaucoma). Increasing the NAM dose combined with gene therapy may prove even more protective.

In conclusion, we show that dietary supplementation with a single molecule (vitamin B_3 or NAM) or *Nmnat1* gene therapy significantly reduces vulnerability to glaucoma by supporting mitochondrial health and metabolism. Combined with established medications that lower IOP, NAM treatment (and/or *Nmnat1* gene therapy) may be profoundly protective. By providing a new molecular and metabolic link between increased neuronal vulnerability with age and neurodegeneration, these findings are of critical importance for glaucoma and possibly other age-related diseases.

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ACKNOWLEDGMENTS

The data reported in this paper are available in the supplementary materials. RNA-seq data are available through the Gene Expression Omnibus (accession number GSE90654). The authors would like to thank the staff of the flow cytometry, histology, gene expression services, and computational sciences at the Jackson Laboratory; G. Howell and R. Libby for discussion and experiment design; K. Kizhatil for reading the manuscript; M. de Vries for assistance with diet and organizing; B. Cardozo for colony maintenance and drug changes; and A. Bell for intraocular pressure measurements. The authors would also like to thank their sources of funding: the Jackson Laboratory Fellowships (P.A.W. and J.M.H.), partial support from EY11721 (S.W.M.J.), the Barbara and Joseph Cohen Foundation (S.W.M.J.), and HL49277 (O.S.). S.W.M.J. is an Investigator of The Howard Hughes Medical Institute. S.W.M.J. and P.A.W. are inventors on a provisional patent application (no current patent number) submitted by the Jackson Laboratory that covers NAD-related therapies in glaucoma. S.W.M.J. holds additional philanthropic funding from the Lano Family Foundation.

SUPPLEMENTARY MATERIALS

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Materials and Methods
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29 September 2016; accepted 23 December 2016
10.1126/science.aal0092

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Science **355** (6326), 756-760. [doi: 10.1126/science.aal0092]

Editor's Summary

Vitamin B_3 protects mice from glaucoma

Glaucoma is the most common cause of age-related blindness in the United States. There is currently no cure, and once vision is lost, the condition is irreversible. Williams *et al.* now report that vitamin B_3 (also known as niacin) prevents eye degeneration in glaucoma-prone mice (see the Perspective by Crowston and Trounce). Supplementing the diets of young mice with vitamin B_3 averted early signs of glaucoma. Vitamin B_3 also halted further glaucoma development in aged mice that already showed signs of the disease. Thus, healthy intake of vitamin B_3 may protect eyesight.

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